INSTITUTIONAL WHITEPAPER · SIX-LAYER DEPIN

# Hardware-verified *truth* for climate markets and AI compute.

A six-layer DePIN architecture pairing edge-bound cryptographic hardware with a continuous, hardware-signed evidence pipeline — across carbon removal, energy markets, parametric insurance, municipal compliance, AI compute accountability, supply chain, and climate-finance clearing.

| STATUS | FRAMEWORK | MAINNET TARGET | CARDANO PREPROD |
|---|---|---|---|
| *Pre-Launch* | Six-Layer DePIN | Q3 2026 | Active |



*Document MLM-WP-v1.0 / Letter / 8.5″ × 11″*

HARDWARE-VERIFIED · CARDANO · HEX NODE

FRONT MATTER · AUTHORS & ABSTRACT

# A six-layer DePIN *for institutional climate markets.*


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ABSTRACT

Mālama Labs presents a decentralized physical infrastructure network (DePIN) that addresses the foundational **Truth Gap** in global environmental markets and high-performance computing clusters. By pairing specialized, edge-bound hardware configurations with cryptographic data validation, the protocol establishes a continuous, hardware-signed evidence pipeline for carbon removal, energy markets, parametric insurance, municipal compliance, AI compute accountability, supply chain, and climate-finance clearing.

The protocol operates across **seven primary verticals**: Carbon Verification (dMRV), Energy-Market Grid Telemetry, Parametric Insurance Triggers, Municipal Compliance Tracking, $LCO_2$ Climate-Finance Clearing, Hardware-Attested AI Compute Accountability, and Supply Chain Scope 3 Ingestion. All seven share a single signing architecture and validator network.

A six-layer architectural stack establishes absolute cryptographic provenance at the point of data capture. To ensure sustainable tokenomics, scheduled emissions follow a smooth eight-year taper (**60.0M MLMA, 12.0% of the 500M hard cap**), with protocol revenue scaling to fund operator and staker distributions by Year 4 to 5. After Year 8, the protocol operates entirely on revenue distribution. Modeled Year 5 recurring oracle and data-licensing revenue is projected at **$41.64M** across the seven verticals.

CONTENTS

# Twenty-five sections. *One signing architecture.*

From the architectural thesis through the six-layer DePIN stack, tokenomics, regulatory posture, and the operational threat-model appendix.

01 · EXECUTIVE SUMMARY

# Decoupling network growth from *open-ended issuance.*

Legacy Decentralized Physical Infrastructure Networks (DePINs) often face structural vulnerabilities driven by token inflation preceding commercial demand. When network incentives drop, node operators capitulate, causing data fragmentation and destroying institutional utility. Mālama Labs corrects this failure mode through a deliberate decoupling of network growth incentives from open-ended token issuance: scheduled emissions are capped at **60.0M MLMA** across an eight-year smooth taper, and the protocol transitions to revenue-funded economics from Year 4 onward.

The network architecture is organized into an institutional **six-layer DePIN model**. While standard decentralized literature details a canonical five-layer layout, Mālama introduces a protocol-specific extension by separating geographic reward scaling and redemption / governance execution into distinct layers. This explicit isolation guarantees granular auditability, system upgradeability, operational execution clarity, and institutional legibility.

Operating on a native utility asset **MLMA** alongside a multi-instrument environmental ledger architecture (**$LCO_2$ / $VCO_2$ / $LCO_2$-AI / $VCO_2$-AI**), the protocol processes raw telemetry into verified, auditable data packages called **SaveCards**. Modeled Year 5 recurring oracle and data-licensing revenue is projected at $41.64M across the seven core verticals.

| HARD CAP | SCHEDULED EMISSIONS | YEAR 5 RECURRING | PILOT SAVECARDS |
|---|---|---|---|
| 500M MLMA | 60M / 12.0% | $41.6 M | 2,786 + |

By deploying hardware-enclosed cryptographic signatures directly to fields, industrial processing centers, and data centers, Mālama moves past speculative infrastructure narratives into a production-grade verification engine for global climate markets.

02 · INTRODUCTION

# The DePIN problem, the accountability gap, and the *seven verticals.*

## 2.1 The Legacy DePIN Failure Mode

The scaling model of early DePIN initiatives focused on device quantity over strategic placement. Hobbyists deployed duplicate hardware inside well-connected suburban homes, resulting in excess density in regions with low commercial data value, while critical data deserts — remote agricultural land, frontier mining operations, underserved urban industrial corridors — remained unmonitored.

Further, legacy models used inflationary token emission structures with no distinct termination horizons. When emissions decreased, speculative interest declined, operators disconnected their equipment, and enterprise buyers were left with fragmented data streams that failed to meet corporate audit and compliance requirements.

## 2.2 The 400 TWh Energy and Carbon Accountability Gap

This data fragmentation is highly pronounced at the intersection of high-performance AI data centers and regional energy markets. Global AI data center electricity consumption is projected to surpass **400 TWh annually**, creating an accountability gap regarding real-time power draw, marginal carbon intensity, and cooling water attribution. A Federation of American Scientists analysis concluded that Meta's actual emissions may be up to **19,000×** higher than market-based reports suggest.

> **● THE SOFTWARE SELF-MEASUREMENT PROBLEM**
>
> Hyper-scale data center operators often rely on generalized monthly utility statements and unverified Renewable Energy Certificates (RECs) to declare carbon neutrality. These backward-looking instruments mask real-time reliance on carbon-heavy peaker plants during periods of high grid demand. Software-based telemetry that runs on the same machines it is measuring cannot resolve this; only hardware-signed measurement at the rack, independent of the host software, can.

## 2.3 The Seven-Vertical Platform Thesis

Mālama Labs replaces unverified data pipelines with a standardized, hardware-verified data collection framework across seven core industrial platform verticals: Carbon Verification (dMRV), Energy-Market Grid Telemetry, Parametric Insurance Triggers, Municipal Compliance Tracking, $LCO_2$ Climate-Finance Clearing, Hardware-Attested AI Compute Accountability, and Supply Chain Scope 3 Ingestion.

By deploying specialized hardware directly to the point of resource consumption or carbon removal, the network transforms physical observation into an auditable digital asset class, resolving tracking challenges for enterprise buyers, carbon registries, and compliance bodies.

03 · THE ARCHITECTURAL THESIS

# From five layers in the literature to *six layers* in Mālama.

## 3.1 The Six-Layer Extension

Academic literature analyzing standard DePIN infrastructure outlines a five-layer layout: Physical, Verification, Storage / Off-Chain, Blockchain, and Application. Mālama explicitly extends this paradigm into a specialized six-layer architecture, separating two functions into independent layers:

- **Reward Scaling Layer (Layer 5)** — handles real-time geographic incentives, density limits, and cell saturation controls via the H3 spatial index (open-source, originally developed by Uber).
- **Redemption Layer (Layer 6)** — controls user claims, vote-escrowed token staking, fee distributions, data retrieval authorizations, and corporate procurement access keys.

## 3.2 Why This Separation Matters

Conflating geographic optimization metrics and treasury distributions within a generalized application layer obscures system risks and reduces modular efficiency. Explicitly separating these functions into independent architectural layers delivers four primary advantages:

01 **Granular auditability.** Enterprise compliance officers, financial regulators, and carbon registries can independently evaluate individual components of the data and value lifecycle.

02 **Targeted upgradeability.** The core data verification or cryptographic stack can be upgraded without disrupting the blockchain settlement layer or altering the governance staking state.

03 **Operational execution clarity.** Isolating hardware-edge mechanics from treasury distribution ensures high-volume cryptographic computation never competes for resources with protocol settlement logic.

04 **Institutional legibility.** Corporate compliance officers can trace how a raw physical measurement in the field translates into an unalterable on-chain settlement asset.

04 · LAYER 1 — PHYSICAL

# Hardware-rooted identity, isolated *at silicon.*

## 4.1 Hardware Specification Status and Design Lineage

The hardware described throughout Sections 4, 10, and 11 represents the current reference design for Mālama's field signing nodes, rack-form-factor signing devices, and associated sensor modules. Final production specifications and manufacturing details remain subject to engineering iteration through Genesis 200 hardware delivery (end of Q4 2026). Specific component selections, supplier identities, and detailed bill-of-materials are maintained under engineering and supplier confidentiality. The architectural functions described in this whitepaper are the public commitments; component substitution within the same functional envelope is preserved as an engineering prerogative.

## 4.2 Field Signing Node Specification

The physical architecture of the network is built on the **Genesis Hex Node**, a modular hardware platform engineered for flexible field deployment. As a Physical Resource Network (PRN) primitive, the node relies on localized geographic density to resist replication by digital adversaries, and on hardware-rooted cryptographic identity to resist data forgery.

Each node contains a tamper-resistant hardware secure element with a non-exportable ECDSA signing key, provisioned at the device at manufacture via an onboard hardware true random number generator. The private key is physically isolated from the file system and the application processor; signing operations are performed inside the secure element, and the key never leaves silicon. This provides an unforgeable hardware identity that cryptographically signs every telemetry packet before it leaves the device.

## 4.3 Interchangeable Sensor Modules

Each Genesis Hex Node features an industrial, weatherproof plug-and-play bus supporting external sensor modules. Two primary configurations:

> ● **WEATHER STATION MODULE**
>
> Engineered for micro-climate tracking, grid optimization, and parametric insurance calibration. Allows operators to bypass infrastructure lease and acquisition overhead typically forced upon centralized weather data providers.
>
> **Tracks:** ambient temperature and relative humidity; barometric pressure and air quality indices; wind speed, direction, and rainfall accumulation.

● SOIL STATION MODULE

Engineered for regenerative agriculture, soil organic carbon monitoring, and verified agricultural land management — mapping directly to standard frameworks like Verra VM0042.

**Multi-probe array monitors:** volumetric water content (soil moisture percentage); soil temperature at staggered root depths; bulk electrical conductivity, pH levels, and primary nutrient dynamics.

## 4.4 Hardware Resilience and Offline Buffering

Each Genesis Hex Node features a local, encrypted non-volatile memory queue managed inside its secure operating layer. If cellular networks or local backhaul experience a complete outage, the unit automatically shifts into offline mode. The hardware continues to log physical observations, execute edge-level cryptographic signing, and track sequence counters for an extended offline operation window. Once backhaul connectivity is restored, the node handles a throttled transmission queue, cross-checking sequence nonces with the public ledger to prevent replay attacks or data duplication.

05 · LAYER 2 — VERIFICATION

# Multilayer pipeline.
# *Five threat classes.*

## 5.1 The Multilayer Verification Pipeline

Data packets transmitted from Layer 1 must pass through an asynchronous verification pipeline before they can be acknowledged by the protocol or written to any storage layer. This pipeline filters out malicious actors, faulty hardware, and anomalous environmental events.

## 5.2 Anti-Spoofing Taxonomy (Class A–E)

To protect the network from coordinated data attacks or signal simulation, the Verification Layer maps incoming packets against five distinct threat classifications. The complete threat matrix is reproduced in **Appendix C**.

This structural verification standard implements an enterprise-grade **Proof-of-Physical-Work (PoPW)** protocol. By enforcing cryptographic validation at Layer 2, the network establishes reliability economics: validators participating in the network do so under a non-transferable PONO credential earned through KYB and operational milestones; falsification or anti-protocol behavior results in PONO revocation and forfeiture of unvested operator rewards. The geographic consensus layer (H3 k-ring BFT attestation) makes single-validator falsification detectable by neighboring nodes.

## 5.3 Validator Ensemble Logic

The network calculates a total data quality score for every received telemetry epoch batch utilizing a weighted multi-variable matrix combining cryptographic, protocol, physical, spatial, temporal, and methodology checks. Each sub-score is normalized dynamically. If an epoch collection fails to clear a strict minimum threshold, the packet is dropped from the active reward loop and flagged for manual system review.

> ● POPW CREDENTIALING — AT A GLANCE
>
> PONO credentials are non-transferable, KYB-gated, and tied to ongoing operational service. Operator vesting is forfeitable. Geographic consensus is by H3 k-ring BFT attestation. Single-validator falsification is detectable by neighboring nodes; cluster falsification is suppressed by spatial neighbor variance correlation (see Class C, Appendix C).

06 · LAYER 3 — ORACLE & OFF-CHAIN

# Decoupling raw telemetry from *execution.*

## 6.1 Data Availability and Scale Logic

To prevent ledger bloat and protect the network from variable transaction fees, raw telemetry streams are decoupled from the execution layer. Detailed sensor payloads are compiled off-chain and archived within decentralized data availability layers (Arweave, Celestia, or equivalent) to guarantee long-horizon retrievability.

To anchor these datasets on public ledgers with constant-time scaling efficiency, the Oracle Layer packages thousands of validated Layer 2 data packets into structured geographic and temporal epochs, constructing a highly secure Merkle Tree.

## 6.2 SaveCard Production Data Primitive

The output of this Merkle aggregation is the **SaveCard**, which acts as the primary evidence asset of the Mālama ecosystem. A SaveCard is a portable, cryptographically sealed, and fully auditable data primitive containing a batch of verified environmental telemetry.

Instead of forcing an auditor to look through billions of unorganized database entries, the SaveCard packages a distinct temporal and geographic window into a standardized object anchored on Cardano via `CIP-68`.

> *SaveCards are the primary evidence asset of the ecosystem — portable, sealed, fully auditable, and anchored on Cardano via CIP-68.*

07 · LAYER 4 — BLOCKCHAIN

# Two chains. *One signing architecture.*

## 7.1 Two-Chain Ledger Architecture

Mālama Labs operates a two-chain ledger architecture, separating archival data custody from token and execution functions:

- **Archival and Custody Layer (Cardano Mainnet).** Cardano's Extended UTXO (EUTXO) design, deterministic gas mechanics, and rich native-token metadata structures host the permanent SaveCard audit trails and international carbon registry connections.
- **Execution and Incentive Layer (Base).** Base, the Coinbase-developed Ethereum Layer 2, delivers the EVM-native execution, deep developer ecosystem, and low transaction friction required to process continuous node reward distributions, MLMA token activity, and interactive veMLMA governance.

## 7.2 Cardano Asset Representation (CIP-68)

SaveCards are minted natively on Cardano utilizing the **CIP-68 token standard**. CIP-68 splits the asset into an unalterable reference token containing the cryptographic data roots and a user-facing token that dictates access permissions. This configuration allows SaveCard metadata states to update through programmatic governance while preserving the immutable evidence record.

## 7.3 Independent Operation, Not Bridged

The two chains operate independently. SaveCards live on Cardano. The MLMA token, veMLMA staking, governance, and reward distribution live on Base. The protocol does not require these chains to share state through a bridge, and no cross-chain bridge is deployed at Token Generation Event (TGE). Operators do not move tokens between chains as part of normal protocol activity.

A Cardano-side MLMA representation is a Phase 2 question contingent on bridge selection, security audit, and market readiness. It is not on the critical path.

08 · LAYER 5 — REWARD SCALING

# The map problem, solved with *H3.*

## 8.1 H3 Geographic Optimization

To prevent hardware clustering in hyper-connected urban locations, the protocol maps the physical surface of the Earth into discrete hexagonal boundaries using the **H3 spatial indexing system** (open-source, originally developed by Uber). The network tracks all reward coefficients at H3 Resolution 5, dividing the planet into approximately **2,016,842 unique cells**, each averaging approximately **252.9 km²** with approximately 8.54 km average edge length. Resolution 5 balances regional granularity against governance manageability.

| H3 RESOLUTION | UNIQUE CELLS | AVG AREA | AVG EDGE |
|---|---|---|---|
| 5 | 2,016,842 | 252.9 km² | 8.54 km |

This separation solves a core structural design flaw defined in modern network architecture as the **map problem**. Generic token incentive designs fail in the physical world when they incentivize duplicate density inside a single affluent neighborhood; Mālama's explicit Layer 5 architecture programmatically corrects this by anchoring rewards to geographic value, not population density.

## 8.2 Real-Time Saturation Multipliers

Baseline node token rewards are modified at the close of every operational epoch to produce a final adjusted distribution using local cell density data. The protocol enforces strict geographic constraints across five primary zone types:

| ZONE TYPE | TARGET DENSITY | NODES / CELL | MULTIPLIER | STRATEGIC INTENT |
|---|---|---|---|---|
| Urban | >1M | 20 | 0.5× | Suppress unnecessary hardware clustering |
| Dense Suburban | 100K – 1M | 5 | 1.0× | Standard baseline operational density |
| Rural | 10K – 100K | 2 | 1.5× | Encourage early expansion into agricultural zones |
| Frontier | <10K | 1 | 2.0× | Strongly incentivize remote climate MRV setup |
| Strategic / Extreme | Uninhabited / Gap | 1 | Up to 3.0× | Direct capital to high-value carbon sinks and islands |

## 8.3 Validation Reward Formula

```
R = Pool × Geographic × Uptime × Genesis × Stewardship × PoolFactor
```

**Pool** per-validator share of the monthly emission pool released under the KPI-scaled emission schedule. **Geographic** 0.5× (urban) to 3.0× (strategic gap). **Uptime** 1.0× baseline to 1.5× at 99.9%+. **Genesis** 1.5× Year 1 only for Genesis 200 operators, 1.0× thereafter. **Stewardship** 1.5× for operators in qualifying community-stewardship hexes. **PoolFactor** normalization ensuring total payouts do not exceed the monthly emission pool. Rewards are competitive and relative to the active validator set, not fixed.

09 · LAYER 6 — REDEMPTION

# Six gateways and a *programmatic waterfall.*

## 9.1 Core Functional Gateways

The Redemption Layer handles enterprise data accessibility and public economic distribution through six core gateways:

01 **Historical SaveCard Retrieval Portal.** Grants authorized compliance auditors or corporate buyers direct access to historical records and data verification keys.

02 **Programmatic MLMA Claiming Engine.** Allows hardware hosts to claim verified data incentives securely on Base.

03 **veMLMA Vote-Escrowed Lockups.** Secures long-term token commitments to grant network voting weight, subject to PONO credential gating and the 10% per-UBO cap.

04 **Decentralized Parameter Adjustment.** Coordinates on-chain consensus votes to adjust baseline reward scaling factors or update firmware validation rules, under tiered governance thresholds.

05 **Live Calibration Analytics.** Logs hardware drift metrics, keeping edge sensors within strict data tolerances.

06 **$LCO_2$ Collateral Management Framework.** Facilitates smart contract verification paths for advanced carbon finance pre-funding models.

## 9.2 Programmatic Revenue Waterfall

All external enterprise capital accumulated via data licensing, verification fees, and oracle operations passes monthly through an unalterable smart contract waterfall. The structure operates in two phases, separated by the 250M circulating-supply burn floor:

| PHASE | BURN | OPERATORS | VEMLMA | FOUNDATION |
|---|---|---|---|---|
| **Pre-Burn-Floor** · until 250M circulating | 45% | 20% | 15% | 20% |
| **Post-Burn-Floor** · after 250M reached | 0% | 20% | 15% | 65% |

The burn mechanism is the structural deflationary feature of the protocol: while emissions expand supply during the cold-start phase, the burn contracts it as revenue scales, producing a long-run supply curve that approaches the 250M circulating floor from below.

10 · PRODUCT CATALOG

# Commercial product catalog and *unit economics.*

To establish institutional alignment between the whitepaper and the companion financial model, the network is commercializing its technology across a structured product catalog. Pricing, margins, and SaaS subscription rates shown below are reference categories rather than locked SKUs.

● NOTE ON SPATIAL BUNDLING

The standalone field signing node hardware is priced at a baseline retail tier. For institutional deployments requiring exclusive regional rights, this hardware is commercialized as part of the **Genesis 200 Hex-Zone Bundle priced at $2,000.00**, which provides the hardware, a non-exclusive geographic operating license for one H3 Resolution 5 cell, and validator registration in the Mālama network. The Genesis 200 bundle does not include a pre-allocated MLMA token grant; operators earn MLMA exclusively through the validation reward formula based on service performance.

| PRODUCT CATEGORY | VERTICAL / APPLICATION | TIER | ANNUAL SAAS |
|---|---|---|---|
| Field Signing Node (Core) | Core signing platform | Entry | Standard |
| ERW Soil Sensor | Enhanced rock weathering | Mid | Standard |
| ERW Aqueous Sensor | Ocean alkalinity tracking | Mid | Enterprise |
| Biochar Process Controller (Basic) | Small pyrolysis control | Mid | Standard |
| Biochar Process Controller (Industrial) | Industrial processing plant | Premium | Enterprise |
| Direct Air Capture Monitor | DAC facility monitor | Enterprise | Enterprise |
| Energy Substation Tracker | Energy grid tracker | Mid | Standard |
| Rack-Mount AI Compute Monitor | AI compute facility | Enterprise | Enterprise |
| Genesis 200 Hex-Zone Bundle | Geographic operating rights | $2,000.00 | Operator (formula) |

11 · AI COMPUTE ACCOUNTABILITY

# Rack-level signing for *hardware-attested* AI compute.

## 11.1 The Rack Attestation Architecture

Mālama's rack-form-factor signing device is a specialized server-rack appliance designed to audit high-performance computing facilities. Rather than assessing data center emissions via historical energy invoices, the device records energy draw at the power distribution unit (PDU) at the rack, signs every record inside a tamper-resistant secure element before release, and anchors the signed record to a public ledger.

The device integrates a high-precision energy metrology IC for real-time voltage, current, and phase tracking. A dedicated application processor running an isolated trusted execution environment hosts the signing pipeline. Every signed packet binds device identity, timestamp, energy reading, and a monotonic counter into a single cryptographic digest before transmission.

● REFERENCE EXEMPLAR · 5-SECOND AI VIDEO CLIP

Hardware-signed measurement of a single short-form generative AI workload demonstrates the order-of-magnitude divergence between software self-reports and physical PDU draw.

| ENERGY | UNDERREPORTED | WATER | GRID CARBON |
|---|---|---|---|
| 944 Wh | 19,000 × | 0.74 L | 412 g/kWh |

## 11.2 Real-Time Emission Flow Oracle Mapping

The rack attestation device functions as an institutional interface to a fungible Digital Resource Network (DRN). While standard DRNs provide decentralized GPU compute layers at heavily optimized cost envelopes compared to legacy centralized cloud providers, they introduce severe tracking challenges regarding marginal carbon emissions. Mālama resolves this by cross-referencing real-time power draw against live spatial emission indices using regional grid APIs (such as ElectricityMaps and WattTime). This process allows data center clusters to map the precise marginal carbon intensity of their operations down to the exact hour and location. Telemetry from the rack attestation device feeds directly into the Mālama Hex Node validator network.

## 11.3 The Green-Grid Cross-Vertical Bridge

The data validated by the rack attestation device is synchronized with grid-meter deployments at adjacent power substations. This creates a multi-point ledger mapping energy generation directly to high-performance compute consumption. Enterprise buyers and auditors can utilize this hardware-linked ledger to verify that AI compute claims are aligned with verified upstream grid generation, closing the trust loop between energy production and energy use.

12 · ENVIRONMENTAL INSTRUMENTS

# Four instruments.
# *One provenance chain.*

Mālama Labs operates a multi-instrument ledger design to separate nature-based carbon sequestration assets from technology-driven energy mitigation records:

| INSTRUMENT | CLASS | WHAT IT REPRESENTS |
|---|---|---|
| $LCO_2$ | Nature-Based Compliance | Physical, permanently sequestered carbon volume generated via biochar deployments or enhanced rock weathering soil applications. Used as pre-finance instrument against the live data stream at a protocol-set discount to expected verified value. |
| $VCO_2$ | Nature-Based Voluntary | Governs voluntary corporate carbon removal credits backed by complete historical SaveCard proofs. Issued upon registry approval and full methodology validation. |
| $LCO_2$-AI | Compute Compliance | Tracks real-time Scope 2 power mitigation offsets paired with rack attestation device telemetry to balance machine learning computational footprints. Designed for institutional counterparty transactions within AI procurement. |
| $VCO_2$-AI | Compute Voluntary | Provides verifiable, audit-grade carbon offset instruments tailored for consumer-facing green AI product assertions. Backed by hardware-signed rack-level evidence. |

All four instruments share a common cryptographic provenance chain back to hardware-signed source data and are subject to the same validator and registry compliance gates.

13 · MARKET SEGMENTATION

# Seven markets. *One signing architecture.*

The multi-vertical commercial structure of Mālama Labs aligns with a broader maturation of the DePIN ecosystem. The global DePIN sector reached a circulating market capitalization surpassing **$10 billion** while demonstrating sustained bear market resilience. Networks now trade at sustainable **10–25× revenue multiples**, indicating a structural shift away from highly speculative legacy retail cycles. The World Economic Forum projects that DePIN protocols will unlock substantial economic value across infrastructure verticals over the next decade.

Mālama monetizes its six-layer architecture across seven primary verticals:

- **Durable Carbon Removal (dMRV)** — automated logging for biochar pyrolysis plants and enhanced rock weathering arrays, cutting registry credit validation timelines from 14 months down to 5 months.
- **Grid Telemetry Analytics (DeGEN)** — high-frequency environmental mapping at regional power nodes. Implements an Energy DePIN model, using crypto-economic incentives to integrate Distributed Energy Resources (DERs) such as solar arrays and smart batteries into virtual power networks.
- **Parametric Financial Settlement** — providing continuous weather verification data used as dispute-free payout triggers for index-based crop and property insurance contracts.
- **Municipal Compliance Tracking** — continuous district air quality, particulate matter, and urban heat island telemetry required for municipal planning and emissions compliance reporting.
- **Data Center Compute Accountability** — linking directly to high-performance computing clusters via rack attestation devices to track power consumption and ambient cooling profiles, auditing real-time carbon emissions claims.
- **Supply Chain Scope 3 Ingestion** — hardware-verified environmental baselines for corporate agricultural supply chains, satisfying international sustainability disclosure requirements.
- **$LCO_2$ Climate-Finance Clearing** — processing pre-finance documentation and verifying cross-border carbon removal assets within specialized digital settlement clearings.

14 · TOKENOMICS

# 500M hard cap. *60M emissions.* Eight years.

## 14.1 Fixed MLMA Supply and Allocation

The MLMA token serves as the core utility, staking, and coordination asset of the network. Total supply is hard-capped at **500,000,000 MLMA**, enforced on-chain via an immutable cap parameter. The hard cap cannot be modified through governance; modification would require a new contract deployment and separate participant consent.

| HARD CAP | EMISSIONS | WINDOW | PUBLIC SALE |
|---|---|---|---|
| 500M MLMA | 60M / 12% | 8 years | 0 at TGE |

### TOP-LEVEL ALLOCATION

| POOL | ALLOC | TOKENS | VESTING STRUCTURE |
|---|---|---|---|
| Investors | 30.0% | 150,000,000 | 12-month cliff + 48-month S-curve |
| Insiders (Team + Advisors) | 20.0% | 100,000,000 | 12-month cliff + 48-month S-curve |
| Foundation / Treasury | 15.0% | 75,000,000 | Governance-directed releases |
| Community | 35.0% | 175,000,000 | Itemized below |
| Public Sale | 0% | 0 | No public sale at TGE |

### COMMUNITY SUB-ALLOCATION

| COMPONENT | ALLOC | TOKENS | FUNCTION |
|---|---|---|---|
| Genesis 200 operators | 5.0% | 25,000,000 | 200 hex zones × 125,000 MLMA, formula-earned |
| Stewardship Pool | 1.75% | 8,750,000 | Indigenous community-led deployments, FPIC-gated |
| Network emissions (Yrs 1–8) | 12.0% | 60,000,000 | KPI-scaled validator rewards, 8-year window |
| Post-emission governance reserve | 16.25% | 81,250,000 | Unreleased after Year 8; governance-directed |

14.2 · EIGHT-YEAR SMOOTH EMISSION TAPER

# From cold-start to *permanent revenue-funded.*

To ensure long-term economic discipline, Mālama emissions are constrained to a smooth eight-year taper rather than an open-ended inflation curve. Total scheduled emissions are capped at **60.0M MLMA (12.0% of supply)**. From Year 9 onward, scheduled emissions are zero and the protocol operates entirely on revenue distribution.

| YEAR | EMISSION CEILING | MONTHLY CEILING | PHASE |
|---|---|---|---|
| Year 1 | 12.0M MLMA | 1.000M / mo | Cold-start (emission-dependent) |
| Year 2 | 14.0M MLMA | 1.167M / mo | Scaling (emission-dependent) |
| Year 3 | 12.0M MLMA | 1.000M / mo | Transitioning |
| Year 4 | 9.0M MLMA | 0.750M / mo | Revenue-majority |
| Year 5 | 6.0M MLMA | 0.500M / mo | Revenue-funded |
| Year 6 | 4.0M MLMA | 0.333M / mo | Revenue-funded |
| Year 7 | 2.0M MLMA | 0.167M / mo | Revenue-funded |
| Year 8 | 1.0M MLMA | 0.083M / mo | Final emission year |
| Year 9+ | 0 | 0 | Permanent revenue-funded |
| **Total** | 60.0M MLMA | | 12.0% of supply |

Monthly release scales with on-chain network growth metrics (validator count, SaveCard count, veMLMA TVL) under a smooth function with a 25% floor and 100% ceiling. The mechanism prevents emission cliffs and discourages threshold-edge gaming.

● HONEST FRAMING OF THE EMISSION DEPENDENCY CURVE

The protocol does not survive emission cessation in Year 1. It survives by Year 4 to 5. Years 1 to 3 require emissions to bootstrap operator economics, which is the standard DePIN cold-start pattern. The transition from emission-dependent to revenue-funded operator economics occurs structurally across the 8-year taper as protocol revenue scales.

15 · REVENUE MODEL

# Recurring oracle fees. *Not asset sales.*

## 15.1 The Core Distinction

To maintain clear institutional presentation, Mālama strictly separates **recurring** protocol oracle and data revenue from **non-recurring** network asset sales. Legacy whitepapers frequently combine one-time hardware or spatial access sales with operational data fees, creating a misleading picture of long-run protocol economics. Genesis 200 Hex-Zone sales ($400,000 aggregate at the Phase 1 cap) are recognized as non-recurring infrastructure revenue at delivery. Validation fees, oracle subscriptions, and settlement fees are recurring.

## 15.2 Year 5 Recurring Revenue Profile

The long-term financial viability of the protocol is driven entirely by high-margin enterprise data licensing and automated oracle settlement fees. Year 5 recurring oracle and data-licensing revenue is projected at **$41.64M** across the six monetized verticals below.

| REVENUE STREAM | YEAR 5 VOLUME | CHARACTERIZATION |
|---|---:|---|
| Carbon Verification (dMRV) | $10.80M | High-Margin Core Recurring |
| Energy-Market Data | $8.06M | High-Margin Core Recurring |
| Parametric Insurance | $1.80M | High-Margin Core Recurring |
| Smart-City Data Licensing | $12.50M | High-Margin Core Recurring |
| Hardware-Attested AI Compute | $7.20M | High-Margin Core Recurring |
| $LCO_2$ Settlement Fees | $1.28M | High-Margin Core Recurring |
| **Total Recurring Pool** | **$41.64M** | **Fully Sustainable Operational** |

Supply Chain Scope 3 Ingestion is included in the seven-vertical commercial platform but folded within general Carbon Verification software licensing at this pre-launch gate, until pilot customer integrations define a distinct per-node reporting line. Conservative case (~40% addressable volume) projects approximately **$17M** Year 5 revenue. Base case (~60–70%) projects **$41.64M**. All figures are internal estimates, not audited projections, and are subject to material variance based on actual market penetration, product deployment, and pricing.

16 · MARKET METHODOLOGY

# Serviceable revenue, *per vertical.*

### 16.1 Why Market Methodology Matters

To achieve institutional credibility, a whitepaper must reject vague, trillion-dollar addressable market claims that ignore operational realities. Revenue modeling must be grounded in precise unit economics, realistic corporate buying patterns, and documented operational friction.

### 16.2 Serviceable Revenue Framework

| SEGMENT | MONETIZED UNIT | COMMERCIAL MECHANISM |
|---|---|---|
| Carbon Verification | Cryptographic verification data access | Fixed infrastructure fee per verified ton of removal via SaveCards |
| Energy-Market Data | Weather Module API streams | Multi-year B2B enterprise data access licensing for utility grid forecasting |
| Parametric Insurance | Weather Module threshold logs | Scaled oracle verification fees built into automated insurance smart contracts |
| Smart-City Monitoring | Localized air and environmental profiles | Annual public-sector data procurement agreements for emissions tracking |
| Hardware-Attested AI | Real-time PDU energy and emission maps | Enterprise SaaS subscription fees per audited server rack footprint |
| $LCO_2$ Settlement | Cross-chain settlement verification | Asynchronous settlement fees on advanced cross-chain climate transactions |
| Supply Chain Scope 3 | Hardware-verified environmental baselines | Pilot-driven; folded into Carbon Verification licensing until distinct line emerges |

17 · LIVE PILOTS & FINANCIAL PROFILE

# 2,786 SaveCards. *Live since June 2024.*

## 17.1 Live Field Operational Status: Texas Pilot Node #1

Mālama Labs rejects theoretical performance assertions by anchoring its methodology in active field data tracking. Since June 2024, the protocol has operated the **Texas Pilot Node #1**, an active agricultural deployment monitoring soil parameters and environmental conditions on a continuous basis.

● TEXAS PILOT NODE #1 · PREPROD · LIVE

| SAVECARDS | LEDGER | SINCE | SIGNED |
|---|---|---|---|
| 2,786 + | Cardano preprod | June 2024 | 100 % |

This milestone confirms edge compute stability, backhaul durability, and script security across multiple seasonal horizons.

## 17.2 Active Financing Profile

The ongoing expansion of the network is backed by an active **$5,000,000 seed funding round** structured at a **$20,000,000 post-money valuation cap** via SAFE with token side letter. These funds are dedicated to accelerating manufacturing of field signing hardware, expanding the core technical development team, and completing comprehensive third-party security audits across the Cardano and Base implementations.

The Genesis 200 Hex Node operator sale opens **Late May 2026**, with hardware delivery scheduled for the end of Q4 2026 and an independent Genesis Hex Sale audit in October 2026 ahead of mainnet activation in Q3 2026.

18 · REGISTRY STRATEGY

# Parallel onboarding to *Puro.earth* and *Isometric.*

## 18.1 Parallel Registry Integration

Mālama's dMRV platform is designed as an implementation layer that works with major registries. The protocol runs parallel onboarding tracks with both **Puro.earth** and **Isometric**. This dual-registry track ensures that data compiled via SaveCards meets the compliance demands of nature-based removals and durable industrial removals simultaneously, preserving methodology flexibility for project developers.

## 18.2 Isometric v2 Amendment Co-Authorship

To advance automated, hardware-native tracking across carbon ecosystems, Mālama is co-authoring a methodology amendment for the Isometric v2 enhanced weathering protocol. This technical amendment introduces rules for edge-enclave data logging, shifting authentication protocols from periodic manual sampling toward continuous, cryptographically-signed observation.

19 · GO-TO-MARKET

# Scarcity allocation, the *delegation flywheel,* and FPIC.

## 19.1 Phase 1: Scarcity Allocation

Initial hardware access is restricted to strategic geographic deployment zones. NFT-HEX rights are allocated to high-value industrial sectors, prioritizing regions with active carbon removal initiatives or acute data shortages. The Genesis 200 release covers 200 hex zones across **five city batches**: Los Angeles, New York, London, Tokyo, and Idaho / Sun Valley (Sun Valley batch is summit-attendee-gated).

## 19.2 Phase 2: The Delegation Flywheel

To accelerate global scale without requiring centralized corporate deployment teams, the network implements an automated **Delegation Economy**. This framework allows capital providers to secure geographic access rights via NFT-HEX assets and delegate physical node maintenance to local field operators.

## 19.3 Free, Prior, and Informed Consent (FPIC)

The integration of Free, Prior, and Informed Consent (FPIC) frameworks and indigenous land stewardship rules represents a critical structural boundary for global network scaling. These protocols are undergoing final optimization and formal review under the leadership of Jeffrey Wise (COO, Hawaiʻi Field Operations and Indigenous Stewardship).

The protocol reserves operational expansion within native land boundaries until these compliance gates are finalized, ensuring that community wealth allocation, governance controls, and data protection rules are verified.

20 · COMPARATIVE ANALYSIS

# How Małama maps against *legacy open architecture.*

| DOMAIN | LEGACY OPEN ARCHITECTURE | MĀLAMA IMPLEMENTATION | ADVANCEMENT |
|---|---|---|---|
| Physical Node Substrate | White-box switches, small-cells, commodity server racks | Field signing nodes with hardware secure element + interchangeable sensor modules | Generalized software container hosting → hardware-enclosed root of trust |
| Core Edge Intelligence | Virtualized RAN components and central network operators | Edge processors with local AI inference and integrated GNSS geographic boundary verification | Programmatic traffic localization and sensory health diagnostics executed at the edge |
| Data Provenance & Control | In-band network telemetry via programmable switches | Multi-tier validation pipeline (Class A–E) generating cryptographic SaveCards | Localized data steering → absolute cryptographic provenance |
| Control Plane Distribution | Central cloud-hosted multi-tenant control plane | Two-chain ledger distributing fast settlement to Base while preserving archival custody on Cardano | Decouples transactional execution from archival data, eliminating ledger bloat |
| Ecosystem Scaling Model | Central cloud management platform driving standardized portals | Tokenomics emissions bootstrap paired with smart contract revenue waterfall | Eliminates permanent infrastructure inflation; node compensation transitions from emissions to revenue |

21 · DECISION GATES & DEADLINES

# Three gateways. *One summit.*

## 21.1 Core Engineering Gateways

Prior to launching the public mainnet, the technical core team must clear three critical engineering milestones:

01 **Security Audit Sign-off.** Complete comprehensive, independent smart contract and cryptographic verification audits across both the Cardano (Aiken / Plutus V3) and Base (Solidity) ledger implementations. Genesis Hex Sale audit scheduled for October 2026.

02 **Hardware Module Stress Tests.** Finish extended environmental stress testing of the physical plug-and-play bus connections across the full operational temperature envelope.

03 **AI Engine Threshold Tuning.** Complete training of the anomaly detection models on expanded pilot datasets to minimize false positives during extreme weather events.

## 21.2 Sun Valley Forum Milestone

The final production-ready compliance and engineering review for the Genesis 200 architecture is scheduled around the **Sun Valley Forum (June 15–18, 2026)**. This event serves as a primary technical review gate, moving the network toward mainnet readiness ahead of the Q3 2026 public launch window.

| AUDIT · GENESIS HEX | SUN VALLEY REVIEW | MAINNET |
|---|---|---|
| Oct 2026 | Jun 2026 | Q3 2026 |

22 · REGULATORY POSITIONING

# Utility design, *not investment.*

## 22.1 Utility Infrastructure Design (MLMA)

The MLMA token functions as an on-chain utility instrument for transaction fee settlement, network staking, and governance. Under the SEC's 2019 digital asset framework, its design focuses strictly on functional utility and programmatic coordination, reducing speculative variables under the Howey test. Final regulatory characterization is under preliminary internal review by Beneficial Technology; this whitepaper is not a legal opinion.

The Genesis 200 operator sale is structured as a hardware-and-license bundle (**$380 hardware + $1,620 non-exclusive geographic operating license, total $2,000**). The bundle does not include a pre-allocated MLMA token grant; operators earn MLMA exclusively through the validation reward formula based on service performance. This structure is designed to strengthen the §13 Howey position by anchoring token receipt to substantial ongoing services.

## 22.2 Data Evidentiary Status (SaveCard)

SaveCards are structured as **verified digital evidence portfolios**, not as investment products. Their legal classification aligns with data audit assets used to demonstrate compliance with international sustainability reporting mandates.

## 22.3 Structured Climate Settlements ($LCO_2$ and $VCO_2$)

The $LCO_2$ and $VCO_2$ instruments function as environmental settlement assets designed for institutional counterparty transactions. Their transferability is managed by smart contract authorization filters, ensuring compliance with cross-border climate-finance regulations and environmental registry rules.

23 · CONCLUSION

# The token serves the protocol, *not the speculation.*

Mālama Labs delivers a hardware-verified infrastructure network that addresses data integrity challenges in global environmental and computing markets. By combining edge-bound cryptography on Mālama field signing nodes with a rigid six-layer architecture, the protocol eliminates reliance on unverified data flows in carbon markets, AI emissions disclosure, parametric insurance, and supply chain attestation.

With an asset-backed model tracking seven core markets, a smooth eight-year emission taper (60.0M MLMA total) transitioning to revenue-funded economics, and live validation from the Texas Pilot Node, Mālama establishes a scalable, sustainable data platform for the global climate economy.

> *The token serves the protocol, not the speculation. The validators serve the network, not the token.*

24 · APPENDICES A & B

# SaveCard payload, *cross-chain state machine.*

## A SaveCard Structural Payload Exhibit

The following JSON document details the standardized structural schema generated by Mālama field signing nodes at the close of every data collection interval, illustrating the integration of hardware identifiers, sensor module states, spatial indexing coordinates, and cryptographic signatures.


```
{
  "savecard_metadata": {
    "version": "1.0",
    "batch_id": "0x7a9c8b2d4e1f6a8b9c0d1e2f3a4b5c6d7e8f9a0b…",
    "epoch_timestamp_start": 1773881041,
    "epoch_timestamp_end": 1773884641
  },
  "hardware_identity": {
    "node_id": "0x00F1A9B2C3D4E5F6A7B8C9D0E1F2A3B4",
    "secure_element_attestation": "ECDSA-signed at silicon",
    "firmware_version": "1.0.0"
  },
  "spatial_index": {
    "h3_resolution": 5,
    "h3_cell_id": "85283473fffffff",
    "average_cell_area_km2": 252.9
  },
  "sensor_payload": {
    "module_type": "Soil Station",
    "readings": {...}
  },
  "cryptographic_signature": "0x..."
}
```


## B Operational Cross-Chain State Machine Flow

Operational flow proceeds through six sequential phases:

01 Edge signing at the field signing node.

02 Edge verification across the Class A–E threat taxonomy.

03 Merkle aggregation in the Oracle Layer.

04 SaveCard issuance on Cardano via CIP-68.

05 Reward computation and distribution on Base under the validation reward formula and revenue waterfall.

06 Governance-controlled redemption of operator claims, veMLMA distributions, and treasury actions.

The chains operate independently; no cross-chain bridge is deployed at TGE.

24C · OPERATIONAL THREAT MODEL MATRIX

# Five classes. *Two systemic vectors.*

The following matrix represents the operational tracking layer utilized by the Verification Layer to actively protect the network from physical, data, and systemic exploits:

| THREAT | VECTOR | DETECTION | MITIGATION |
|---|---|---|---|
| Class A<br>RF / GPS SPOOFING | Forging location data to capture strategic multipliers | Multi-tower cellular GNSS timing delta checks | Reject packet; flag H3 cell; route unit to manual review queue |
| Class B<br>STREAM INJECTION | Simulating sensor values into transmission lines | Hardware signature checks via secure element | Instant drop of unsigned data; blacklist unauthorized public keys |
| Class C<br>NEIGHBOR SPOOFING | Deploying multiple nodes to simulate wider coverage | H3 spatial neighbor variance correlation algorithms | Scale down local cell rewards; PONO revocation and vesting forfeiture |
| Class D<br>HISTORICAL REPLAY | Re-transmitting historic high-quality data streams | Tracking strict monotonic sequence nonces on the ledger | Invalidate packet epoch; expire associated transaction nonces |
| Class E<br>ENVIRONMENTAL CLONING | Artificially modifying local node environments | Localized AI edge plausibility and trend-drift filters | Flag data anomalies; require physical device calibration tests |
| Sybil Deployment | Virtualizing nodes via cloud software templates | Verified hardware provisioning policy ID handshake filters | Deny registration to units lacking physical factory keys; KYB-gated PONO credentials |
| Outlier Data Flooding | Overloading validators with high-frequency telemetry | Hard data ingestion rate limits applied per H3 index | Throttle connection endpoints; isolate suspect node IDs |

25 · REFERENCES & INDEXED MACRO CONTEXT

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DISCLAIMER

This document describes the working design of the Mālama protocol and the MLMA utility token. It is not an offer to sell, a solicitation to buy, or a representation of regulatory status in any jurisdiction. All financial projections are internal operating plan estimates, not audited figures. Hardware specifications are reference designs; final production specifications are subject to engineering iteration. MLMA regulatory classification is under preliminary internal review by Beneficial Technology; the Howey analysis is a preliminary internal working draft and must not be relied upon for any regulatory determination. Participation in the Genesis 200 program is subject to KYB requirements, applicable law, and the executed Hex Node Purchase and Preorder Agreement.